GeOsm

Complete user guide

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1. Map and navigation

Interactive map view

GeOsm displays an interactive map built on OpenLayers, with smooth navigation via mouse, trackpad or touch (zoom, pan, rotate). The current position and zoom level are preserved when switching tools, so you never lose track of what you were exploring.

> **Example :** You explore the Bastos neighborhood in Yaoundé by progressively zooming in from the country view down to street level, without ever reloading the page.

Navigation history

Just like a web browser, GeOsm keeps a history of visited positions and zoom levels. The toolbar's back and forward buttons let you quickly return to a previously explored view without having to find it again manually.

> **Example :** After zooming into three different cities to compare their school infrastructure, you return to the first view in one click using the back button.

Context menu (right-click)

Right-clicking (or long-pressing on mobile) anywhere on the map opens a context menu with the actions available at that exact spot: copy coordinates, center the view, start a measurement, create a geosignet, and more.

> **Example :** You right-click on an intersection to instantly copy its GPS coordinates and send them to a colleague in the field.

Zoom to GPS coordinates

If you already know the exact coordinates of a place (for example recorded with a field GPS unit), you can enter them directly so the map centers and zooms in instantly, without going through a name search.

> **Example :** A field agent receives the coordinates "3.8480, 11.5021" by text message and enters them directly to locate the exact point on the map.

Geosignets (saved positions)

A geosignet saves a map position (center and zoom level) under a name of your choice, so you can return to it in one click from any future session, without having to navigate or search for the place again.

> **Example :** You regularly check the Bonapriso neighborhood: you save it as a "Bonapriso" geosignet and find it instantly on every visit.

My Maps (layer compositions)

"My Maps" lets you save a specific combination of active layers (with their opacity and order) under a name, so you can find and reactivate it in one action later, instead of turning on each layer one by one.

> **Example :** You build a "School infrastructure" map combining primary schools, colleges and universities, and recall it in one click at every follow-up meeting.

Map settings

The Settings panel gathers the map's display preferences: interface language (French, English, Spanish), units of measurement, and other options that apply instantly without reloading the page.

> **Example :** An English-speaking user switches the interface to English from the Settings panel in a single action.

Swipe map comparison

The swipe comparison tool overlays two base maps (for example a recent satellite view and an older one) with a slider you drag to progressively reveal one or the other, useful for observing changes in the terrain.

> **Example :** You compare a 2015 satellite image with a 2024 one to visualize urban expansion in a Douala neighborhood.

Descriptive sheet and information popup

Clicking on a map object displays a popup with its main information; a "learn more" button opens an enriched descriptive sheet with links to Wikipedia, Wikidata and the corresponding OpenStreetMap entry when available.

> **Example :** You click on a historic landmark and go straight to its Wikipedia page to learn more about its history.

Tasks tray and notifications

The bell icon in the top bar tracks long-running tasks started from the app in real time (export, location plan, AI analysis report): progress keeps updating even if you switch tools or navigate elsewhere, and a download link appears as soon as the result is ready.

> **Example :** You start a large export and close the panel to keep exploring the map; a few minutes later, the tasks tray shows the file ready to download without you having had to watch its progress.

2. Search and discovery

Geographic search (Nominatim)

The search bar lets you find an address, place name or point of interest by name, powered by the Nominatim geocoding engine. The map automatically centers on the first relevant result.

> **Example :** You type "Marché Central Yaoundé" into the search bar and the map immediately centers on its exact location.

Personalized suggestions

As soon as the map opens, GeOsm proposes a list of suggested layers, computed from the layers most frequently activated by all users of the instance, saving you time browsing the whole catalog.

> **Example :** When opening the Cameroon map, you immediately see "Hospitals", "Schools" and "Main roads" suggested, the most consulted layers.

Hierarchical layer catalog

The catalog organizes all available layers into a Groups > Sub-groups > Layers tree (for example Health > Hospitals), making it easy to navigate even when dozens of layers are available.

> **Example :** You're looking for transport data: you expand the "Transport" group then the "Roads" sub-group to find the exact layer you need.

Active layers (opacity, heatmap, resync)

Once a layer is activated, a dedicated panel lets you adjust its opacity, switch to a heatmap representation to visualize concentrations, and resync it if the source data has been updated on the admin side.

> **Example :** You lower the opacity of the "Administrative boundaries" layer to 40% to better see the satellite base underneath, then switch the "Hospitals" layer to heatmap view to spot underserved areas.

Base map switching

The base map selector lets you switch between several base styles (standard, satellite, terrain...) without losing active layers or the current position.

> **Example :** You switch from the "Standard" base map to "Satellite" to visually check whether a drawn road matches an existing track.

Legend

The legend shows, for each active layer, the meaning of the colors, symbols and icons used, updating automatically as you activate or deactivate layers.

> **Example :** After activating the "Land use" layer, you check the legend to understand what each shade of green represents.

Layer recommendations (co-activation)

In addition to the initial suggestions, GeOsm proposes complementary layers to the ones already active, based on layer combinations frequently activated together by other users.

> **Example :** You activate the "Hospitals" layer and GeOsm suggests "Pharmacies" and "Health centers", often consulted alongside it.

3. Analysis and production tools

Drawing (points, lines, polygons)

The drawing tool lets you sketch points, lines and polygons directly on the map, with the drawing saved so you can find it later or share it with other users of the instance.

> **Example :** A municipal official draws the outlines of flood-prone areas in his neighborhood to share them with his technical team.

Measurement (distance and area)

The measurement tool computes, in real time, the distance of a path or the area of a polygon drawn directly on the map, without needing an external GIS tool for a quick estimate.

> **Example :** You trace the boundary of a plot of land to quickly estimate its area before a field visit.

Routing (OSRM)

Route calculation relies on the OSRM engine to propose a real road route between two points, with distance, estimated duration and the path displayed directly on the map.

> **Example :** You trace a route from Douala to Yaoundé and get a distance of 240 km, an estimated duration and the full path on the map.

Nearest search by real road distance

Rather than relying on straight-line distance, this search finds the nearest entity (a hospital, a school...) accounting for the actual road network, avoiding errors when two geometrically close places are actually separated by an obstacle.

> **Example :** You click on the map to find out which hospital is actually closest by travel time, when two hospitals seem equidistant as the crow flies but one is separated by a river with no bridge.

Spatial analysis (buffer, intersection, union, difference)

The spatial analysis module lets you generate a buffer zone around an object, or combine several geometries via intersection, union or difference, to answer land-use planning questions.

> **Example :** An urban planner creates a 500 m buffer zone around a school to identify all buildings in its immediate vicinity.

Multi-format export

Layer data can be exported in several standard formats (including Shapefile and GeoJSON) for reuse in a desktop GIS tool like QGIS.

> **Example :** A researcher downloads Cameroon's school data in Shapefile format to analyze it in QGIS Desktop.

My data (personal import)

The "My data" tool lets you import your own geographic file (GeoJSON, zipped Shapefile...) to view it as a personal layer, visible only to you; it can then be proposed to the instance's shared catalog, subject to an administrator's review.

> **Example :** A field agent imports a GeoJSON file of GPS points collected during a mission to overlay them immediately on the official layers, without waiting for an administrative publication.

Printing

The print function generates a printable version of the current map view, with active layers, for offline use or inclusion in a report.

> **Example :** You print the current view showing a neighborhood's schools for inclusion in a paper report for a community meeting.

PDF location plan

The location plan generates a professionally laid-out PDF document (A4 format) centered on a chosen point, with a title, a landmark and the surrounding cartographic context, ideal for reaching a specific site in the field. The legend, scale bar, coordinate grid and north arrow can each be toggled independently depending on the level of detail needed.

> **Example :** A field agent picks a point on the map, enters a title and a landmark, turns off the grid for a cleaner layout, then generates an A4 location plan to print for reaching the site.

Geolocated comments

Comments let you pin a remark, a report or a question directly to a point on the map; each pin opens a full discussion thread, with the ability to reply and mark the topic as resolved.

> **Example :** An editor replies to a pothole report to say that repairs are underway, then marks the comment as resolved once the work is done.

Elevation

Clicking on the map shows the elevation of the selected point (from SRTM data); for a traced path, a full elevation profile shows the grade changes along the route.

> **Example :** A road engineer traces a planned road and views the elevation changes along the route to anticipate steep sections.

Mapillary (street-level imagery)

The Mapillary integration displays, where available, street-level panoramic photos at the selected map location, to get a concrete view of a place without traveling there.

> **Example :** Before heading to a site, you check the Mapillary view of an intersection to spot the surrounding shops.

Layer statistics (extent-aware)

The Statistics panel toggles between "Whole layer" and "Currently visible area": charts adapt to each attribute's type (min/max/mean for a numeric attribute, breakdown for a categorical one) and show the total area or length of the selection. The accompanying AI answer doesn't just list numbers - it compares them (visible-area density against the layer's average, for example) to draw a real conclusion.

> **Example :** Zooming into the Douala 5 district, the panel automatically recomputes the number of schools in the visible area and the AI flags that the density there is below the instance average.

Raster statistics (global, by zone, drawn zone)

For a raster layer (a population layer, for example), three modes are available: global statistics, statistics by administrative zone (with a level selector), or statistics over a freely hand-drawn zone. The value shown when clicking a pixel is given in context (estimated inhabitants in the cell and density per km², not a plain raw number).

> **Example :** You hand-draw the outline of a neighborhood with no official administrative boundary and instantly get a population estimate for it, without waiting for an official boundary to be digitized.

Multi-layer analysis and AI report

As soon as two or more layers are active, a dedicated tab in the Statistics panel lets you launch an AI cross-analysis (population, hospitals and schools together, for example) that produces a real deduction rather than a plain list. The "Generate PDF report" button turns that same analysis into a downloadable document, with an "Analyzed area" page showing the place name and a real basemap so the area is instantly recognizable, plus a data-distribution chart per layer.

> **Example :** With Hospitals, Population and Schools active over Bastos, the cross-analysis reports the number of inhabitants per hospital in the area and flags below-average health coverage; the matching PDF report is generated in one click to attach to a file.

My reports (history)

The "My reports" section of the Statistics panel keeps a history of all your generated analysis reports, with direct download from the list; unlike the tasks tray (which only tracks events received while the tab is open), this history stays accessible even after logging out or when a report finished while you were away.

> **Example :** You generate a report on Friday then close your session; the following Monday, you find it again and download it directly from "My reports" without needing to regenerate it.

4. AI Assistant

Natural language queries

The conversational assistant understands questions asked in natural language (in both French and English) about the map's data and features, without requiring you to know any particular syntax.

> **Example :** You type "How many hospitals are there around Yaoundé?" and get a direct answer in natural language.

Assistant-driven map actions

Beyond answering, the assistant can act directly on the map (activate a layer, zoom to an area, run a spatial analysis) thanks to Gemini function-calling, turning a conversation into a real remote control for the map.

> **Example :** You type "Show me the hospitals within 5 km of downtown and zoom in"; the assistant activates the Hospitals layer, runs a proximity analysis, and automatically zooms to the result.

Chained geometry operations

The assistant can compute a buffer, intersection, union or difference directly from the conversation, and remembers the geometries it has created from one message to the next so they can be reused without describing them again.

> **Example :** You ask "Draw a 4 km circle around Douala 5 and another around Douala 3, compute their intersection, then tell me how many hospitals are in it" in a single sentence; the assistant chains all three steps without asking for clarification.

Cross-analysis of the displayed context

On request, the assistant can analyze every layer currently active over the area shown on screen and draw cross-layer conclusions (inhabitants-per-facility ratio, apparent coverage...), exactly like the Statistics panel's multi-layer tab, but directly within the conversation.

> **Example :** With Hospitals and Population active, you ask "Is this area well covered for healthcare?" and the assistant answers by crossing both layers instead of describing each one separately.

AI analysis report generation via chat

You can ask the assistant to write a full report on a given topic from the active layers; generation happens in the background (the chat stays usable meanwhile) and the finished document appears in the tasks tray with a download link.

> **Example :** You ask "Write me a full report on healthcare coverage relative to population in this area"; a few dozen seconds later, the PDF is ready in the tasks tray.

Location plans drafted via chat

From the conversation with the assistant, you can request the creation of a location plan; the AI can also automatically draft the description and landmark when you only provide the title and coordinates.

> **Example :** A field agent in a hurry enters just a title and coordinates, checks "Auto-draft", and lets the AI propose a description and landmark that he corrects before submitting.

Narrated summaries and statistics

The assistant can summarize the current map view or a layer's statistics in a single sentence, handy for a quick report without having to interpret raw figures yourself.

> **Example :** You click "Summarize view" after navigating the map and get a paragraph describing what you're looking at, useful for a quick report or an email.

Conversation history

Every conversation with the assistant is kept and linked to your user account, so you can resume a previous exchange or refer back to it later without retyping everything.

> **Example :** A week later, you find the exchange in which the assistant helped you locate schools in a specific neighborhood.

5. Sharing

Social media sharing

A share button lets you publish the current map view directly to social media, to quickly spread geographic information to a wide audience.

> **Example :** You share a view showing the progress of new road works on social media.

Read-only shared maps

A share link gives access to a specific map view (active layers, position) in read-only mode, viewable by anyone without needing to create an account.

> **Example :** You share a view showing Yaoundé's schools and hospitals with a colleague via a short link, which he views without needing to log in.

6. Account management

Classic login (email/password)

Email and password login remains available to every user. At sign-up, a verification email is sent (the account stays usable in the meantime, with some features limited); if forgotten, the "Forgot password" page sends a reset link by email. The "Remember me" checkbox on the login screen extends the session length so you don't have to log in again on every visit.

> **Example :** You log in with your work email and password, check "Remember me" on a personal computer, to access editing features reserved for editors without re-entering your credentials the next day.

One-click login via OpenStreetMap (OAuth 2.0)

An OpenStreetMap contributor already used to their osm.org account can log in with one click via "Log in with OpenStreetMap", without creating a new password; a local GeOsm account is automatically created on first login.

> **Example :** An OSM contributor clicks "Log in with OpenStreetMap" and immediately accesses the geoportal without going through a classic sign-up.

Profile (including linked OpenStreetMap profile)

The profile panel shows your personal information and, if an OpenStreetMap account is linked, its display name, creation date and number of contributions, with the ability to link or unlink this account at any time.

> **Example :** You check your profile to see how long your OpenStreetMap account has existed and how many contributions are linked to it.

7. Administration

Instance (country) management

Restricted to authorized accounts, instance management lets you create, configure and administer a GeOsm deployment for a given country, with its own layers, users and themes.

> **Example :** The super administrator deploys GeOsm for a new African country, Togo, by defining its coordinates and instance settings.

Theme and layer management

Administrators and editors can create and organize themes (groups and sub-groups), as well as create, style and update the data layers attached to them. A guided wizard offers three ways to create a layer - importing a file, extracting OpenStreetMap data by area/tags, or publishing layers from an existing QGIS project - with a color and icon choice from a predefined catalog in all three cases.

> **Example :** The editor uses the layer creation wizard, picks the "QGIS project" source, selects which layers to publish from an existing project, then assigns them an icon and color before confirming.

User and role management

Authorized accounts can create user accounts without going through public sign-up, and assign the appropriate role (Super Administrator, Instance Administrator, Editor, Viewer) according to each person's responsibilities.

> **Example :** The super administrator creates an account for a new instance manager and assigns them the Instance Administrator role.

A broader back-office

Beyond the instances, themes, layers and users covered above, the back-office (restricted to authorized accounts) also includes a dashboard, moderation of comments and user feedback, review of AI-generated FAQs, background job monitoring and technical supervision tools - outside the scope of this guide, aimed at public geoportal users.

> **Example :** An instance administrator checks the back-office dashboard to see their instance's overall activity at a glance.

8. Help and resources

Info panel

The "i" button in the top bar opens a panel gathering development credits, a link to the open-source code repository, this guide's PDF download (French and English), and a feedback form.

> **Example :** A new user opens the Info panel on their first visit to download the complete guide before starting to explore the map.

Feedback and suggestions

The Info panel's feedback form lets you report an issue, suggest an improvement, or ask a question directly to the platform's administrators, with an optional contact field to receive a reply.

> **Example :** You notice a layer showing incorrect data and report it via the feedback form in a few seconds, without needing to find a contact email.

Open source project

GeOsm is an open-source project: the repository is publicly accessible from the Info panel, and technical contributions from the community are welcome.

> **Example :** A developer interested in the project browses the public repository from the Info panel to explore the code before proposing a contribution.